Complex Numbers: De Moivre's Theorem

Questions 1–24

Question 1. Use De Moivre's Theorem to simplify each of the following, giving your answer in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$:

a) $\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)^{7}$

b) $\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)^{5}$

c) $\left[2\left(\cos\frac{\pi}{5} + \mathrm{i}\sin\frac{\pi}{5}\right)\right]^{11}$

d) $\left[3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)\right]^{8}$

e) $\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)^{-7}$

f) $\left[2\left(\cos\frac{4\pi}{9} + \mathrm{i}\sin\frac{4\pi}{9}\right)\right]^{-4}$

Question 2. Use De Moivre's Theorem to simplify each of the following, giving your answer in the form $a + b\mathrm{i}$ where a and b are real constants:

a) $\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)^{15}$

b) $\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)^{10}$

c) $\left[4\left(\cos\frac{3\pi}{8} + \mathrm{i}\sin\frac{3\pi}{8}\right)\right]^{6}$

d) $\left[\mathrm{i}\left(\cos\frac{5\pi}{4} + \mathrm{i}\sin\frac{5\pi}{4}\right)\right]^{5}$

e) $\left[-3\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)\right]^{6}$

f) $\left[2\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right)\right]^{-4}$

Prerequisite knowledge for Q3: Exponential Form

Question 3. Use De Moivre's Theorem to simplify each of the following, giving your answer in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) $\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)^7$

b) $\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{10}$

c) $\left[3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)\right]^5$

d) $\left[3\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)\right]^8$

e) $\left[2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^{-4}$

f) $\left[-3\left(\cos\frac{5\pi}{8} + i\sin\frac{5\pi}{8}\right)\right]^{-3}$

Question 4. Use De Moivre's Theorem to simplify each of the following leaving your answer in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) $(1 + \sqrt{3}i)^7$

b) $(1 - i)^{10}$

c) $(2i)^{20}$

d) $(-8 + 8\sqrt{3}i)^5$

Question 5. Find the value of

$$\frac{1}{(-1+\sqrt{3}\mathrm{i})^4}$$

giving your answer in the form $a+b\mathrm{i}$ where a and b are non-zero real values.

Question 6. Find the value of

$$\frac{1}{(2-2\mathrm{i})^5}$$

giving your answer in the form $a+b\mathrm{i}$ where a and b are non-zero real values.

Question 7.

a) Write $\cos\theta - \mathrm{i}\sin\theta$ in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$

b) Hence simplify $(\cos\theta - \mathrm{i}\sin\theta)^8$

Question 8. Write

$$\left(\cos\frac{\pi}{5} - \mathrm{i}\sin\frac{\pi}{5}\right)^9$$

in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

Question 9. Without using an expansion, simplify

a) $(1+\mathrm{i})^8 + (1-\mathrm{i})^8$

b) $(\sqrt{3}+\mathrm{i})^7 + (\sqrt{3}-\mathrm{i})^7$

Question 10. Use De Moivre's theorem to simplify

$$(\cos 9\theta + \mathrm{i}\sin 9\theta)(\cos 6\theta + \mathrm{i}\sin 6\theta)$$

Question 11. Use De Moivre's Theorem to write each of the following in the form $r(\cos\theta+\mathrm{i}\sin\theta)$ where $-\pi<\theta\leq\pi$

a) $\left(\cos\frac{4\pi}{3}+\mathrm{i}\sin\frac{4\pi}{3}\right)\left(\cos\frac{7\pi}{3}+\mathrm{i}\sin\frac{7\pi}{3}\right)$

b) $\left(\cos\frac{17\pi}{4}+\mathrm{i}\sin\frac{17\pi}{4}\right)\left(\cos\frac{5\pi}{4}+\mathrm{i}\sin\frac{5\pi}{4}\right)$

c) $\left(\cos\frac{7\pi}{2}+\mathrm{i}\sin\frac{7\pi}{2}\right)\left(\cos\frac{13\pi}{2}-\mathrm{i}\sin\frac{13\pi}{2}\right)$

Question 12.

$$z=\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)^4\left(\cos\frac{\pi}{3}-\mathrm{i}\sin\frac{\pi}{3}\right)^5$$

a) Express z in the form $r(\cos\theta+\mathrm{i}\sin\theta)$ where $-\pi<\theta\leq\pi$

b) Hence find the value of z

Question 13. Use De Moivre's theorem to simplify

a) $\dfrac{(\cos 9\theta + \mathrm{i}\sin 9\theta)^4}{(\cos 4\theta + \mathrm{i}\sin 4\theta)^6}$

b) $\dfrac{(\cos 5\theta + \mathrm{i}\sin 5\theta)^8}{[2(\cos 3\theta + \mathrm{i}\sin 3\theta)]^6}$

Question 14. Simplify each of the following, giving your answer in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$:

a) $\dfrac{\cos\frac{4\pi}{11} + \mathrm{i}\sin\frac{4\pi}{11}}{\cos\frac{5\pi}{11} - \mathrm{i}\sin\frac{5\pi}{11}}$

b) $\dfrac{\left(\cos\frac{9\pi}{14} + \mathrm{i}\sin\frac{9\pi}{14}\right)^6}{\left(\cos\frac{3\pi}{14} - \mathrm{i}\sin\frac{3\pi}{14}\right)^4}$

Question 15. Use De Moivre's theorem to show that for real values a and b, $(a + b\mathrm{i})^n + (a - b\mathrm{i})^n$ is real for all integers n

Question 16. Find the smallest positive integer value of n, such that $\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)^n$ is a real number.

Question 17.

$$z = \left[3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right)\right]^n$$

a) Find the smallest positive integer value of n, such that z is a real number.
b) State the value of z for the smallest positive value of n, such that z is a real number.

Question 18.

$$z = \frac{\sqrt{3} - \mathrm{i}}{\sqrt{3} + \mathrm{i}}$$

a) Express z in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$
b) Find the smallest positive integer value of n such that z^n is a real number.

Question 19.

a) Express $2 + 2\mathrm{i}$ in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$

b) Find the smallest positive integer value of n such that $(2 + 2\mathrm{i})^n$ is a purely imaginary number.

Question 20. Given, for integers m and n,

$$z = \frac{\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)^m}{\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)^n}$$

a) Find the smallest positive integer value of n for which z is purely imaginary, and state the corresponding value of m

b) Find the smallest positive integer value of n for which z is real, and state the corresponding value of z

Question 21. The cubic equation $x^3 - px^2 + qx - 4 = 0$ where p and q are positive integers, has roots $\alpha = 1 + \mathrm{i}$, β and γ

a) Find the values of p and q

b) Show that if n is an integer, then

$$\alpha^n + \beta^n + \gamma^n = 2^{\frac{n}{2}+1} \cos\left(\frac{n\pi}{4}\right) + 2^n$$

Question 22. The cubic equation $x^3 - px^2 + 12x - q = 0$ where p and q are positive integers, has roots $\alpha = 1 + \sqrt{3}\mathrm{i}$, β and γ

a) Find the values of p and q

b) Show that if n is an integer, then

$$\alpha^n + \beta^n + \gamma^n = 2^{n+1} \cos\left(\frac{n\pi}{3}\right) + 2^{2n}$$

Prerequisite Knowledge for Q23: Hyperbolic Functions

Question 23.

a) For integer n, show that

$$(\cosh z + \sinh z)^n = \cosh nz + \sinh nz$$

b) Hence determine the exact value of $\cosh(3\ln 2) + \sinh(3\ln 2)$

Question 24. Given that $z = \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}$

a) Write z^2 and z^3 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$

b) Sketch each of

$$1,\ 1+z,\ 1+z+z^2$$
$$\text{and } 1+z+z^2+z^3$$

on an Argand diagram.

c) State the exact argument of

i) $1+z$

ii) $1+z+z^2$

Complex Numbers: Exponential Form

Questions 1–25

Question 1. Write each of the following in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$:

a) $3\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$
b) $5\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$
c) $2\mathrm{e}^{-\frac{4\pi}{3}\mathrm{i}}$
d) $7\mathrm{e}^{\frac{18\pi}{5}\mathrm{i}}$
e) $-4\mathrm{e}^{\frac{3\pi}{4}\mathrm{i}}$
f) $-2\mathrm{e}^{\frac{5\pi}{4}\mathrm{i}}$

Question 2. Write each of the following in the form $a + b\mathrm{i}$ where a and b are exact real constants:

a) $2\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$
b) $3\mathrm{e}^{\pi\mathrm{i}}$
c) $6\mathrm{e}^{\frac{7\pi}{3}\mathrm{i}}$
d) $2\mathrm{e}^{-\pi\mathrm{i}}$
e) $-4\mathrm{e}^{\frac{5\pi}{2}\mathrm{i}}$
f) $-\sqrt{2}\mathrm{e}^{-\frac{4\pi}{3}\mathrm{i}}$

Question 3. Write each of the following in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$, where r is given as an exact value:

a) $4\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$
b) $5\left(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})\right)$
c) $\sqrt{2}\left(\cos(-\frac{4\pi}{5}) + i\sin(-\frac{4\pi}{5})\right)$
d) $5\left(\cos(-\frac{3\pi}{2}) + i\sin(-\frac{3\pi}{2})\right)$
e) $4\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$
f) $\sqrt{3}\left(\cos\frac{6\pi}{7} - i\sin\frac{6\pi}{7}\right)$
g) $\left(\cos(-\frac{\pi}{5}) - i\sin(-\frac{\pi}{5})\right)$
h) $8\left(\cos(-\frac{5\pi}{4}) - i\sin(-\frac{5\pi}{4})\right)$

Question 4. Write each of the following in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$, where r is given as an exact value, and θ is given as an exact value or rounded to three significant figures if appropriate:

a) $z = 2 + 2i$
b) $z = 4 - 4\sqrt{3}i$
c) $z = -2 + \sqrt{2}i$
d) $z = -5 - 12i$
e) $z = -7$
f) $z = 6i$
g) $z = 8$
h) $z = 3 + i$
i) $z = 1 - \sqrt{5}i$

Question 5. Write down the complex conjugate, z^*, of each of the following, giving your answer in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$.

a) $z = 2e^{\frac{\pi}{5}i}$
b) $z = 8e^{\frac{3\pi}{2}i}$
c) $z = -3e^{\frac{\pi}{7}i}$
d) $z = -6e^{\frac{16\pi}{7}i}$

Question 6. Find each of the following in the form $a + bi$ where a and b are real numbers. Use exact values for a and b where possible or give answers to three significant figures otherwise.

a) $e^{\frac{\pi}{4}i} \times e^{\frac{\pi}{2}i}$
b) $2e^{\frac{\pi}{3}i} \times \sqrt{3}e^{\frac{\pi}{5}i}$
c) $e^{2i} \times -2e^{5i}$
d) $e^{\frac{\pi}{4}i} \times e^{-\frac{\pi}{2}i} \times e^{\frac{\pi}{3}i}$
e) $2e^{-\frac{\pi}{6}i} \times \sqrt{2}e^{\frac{\pi}{3}i} \times e^{\frac{5\pi}{4}i}$

Question 7. Find each of the following in the form $a + b\mathrm{i}$ where a and b are real numbers. Use exact values for a and b where possible or give answers to three significant figures otherwise.

a) $\frac{\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}}{\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}}$

b) $\frac{2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}}{6\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}}$

c) $\frac{\mathrm{e}^{3\mathrm{i}}}{5\mathrm{e}^{7\mathrm{i}}}$

d) $\frac{2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}}{\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}} \times 2\mathrm{e}^{-\frac{\pi}{2}\mathrm{i}}$

e) $\frac{\sqrt{2}\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}}{3\mathrm{e}^{\frac{\pi}{5}\mathrm{i}}} \times \mathrm{e}^{\frac{4\pi}{7}\mathrm{i}}$

Question 8. Write $(\cos 3\theta + \mathrm{i}\sin 3\theta)(\cos 2\theta + \mathrm{i}\sin 2\theta)$ in exponential form.

Question 9. Write each of the following in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$.

a) $\left(\cos\frac{3\pi}{7} + i\sin\frac{3\pi}{7}\right)\left(\cos\frac{4\pi}{7} + i\sin\frac{4\pi}{7}\right)$

b) $\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)\left(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})\right)$

c) $3\left(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})\right)\left(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}\right)$

d) $4\left(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})\right)\left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right)$

Question 10. Write

$$\frac{\cos 6\theta + i\sin 6\theta}{\cos 4\theta + i\sin 4\theta}$$

in exponential form.

Question 11. Write each of the following in the form $e^{i\theta}$, where $-\pi < \theta \leq \pi$:

a) $\frac{\cos\frac{2\pi}{5}+i\sin\frac{2\pi}{5}}{\cos\frac{4\pi}{5}+i\sin\frac{4\pi}{5}}$

b) $\frac{\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}}{\cos(-\frac{2\pi}{3})+i\sin(-\frac{2\pi}{3})}$

c) $\frac{\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}}{\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}}$

Question 12. Two complex numbers are given by $z = 2 - 2i$ and $w = 1 + \sqrt{3}i$

a) Write z in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$

b) Write w in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$

c) Hence show that if $z^2 + w^2 = a + bi$ where a and b are real constants, then

$$\frac{b}{a} = 4 - \sqrt{3}$$

Question 13. Given that $z = 3\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$ and $w = \sqrt{2}\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$, find the value of each of the following:

a) $|zw|$
b) $\left|\frac{z}{w}\right|$
c) $\arg(zw)$
d) $\arg\left(\frac{z}{w}\right)$

Question 14. Given that $z = \sqrt{3}\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}$ and $w = 3\mathrm{e}^{\frac{\pi}{5}\mathrm{i}}$, find the value of each of the following:

a) $|zw|$
b) $\left|\frac{z}{w}\right|$
c) $\arg(zw)$
d) $\arg\left(\frac{z}{w}\right)$

Question 15. Given that $z_1 = r_1 e^{\theta_1 i}$ and $z_2 = r_2 e^{\theta_2 i}$, show that:

a) $|z_1 z_2| = |z_1||z_2|$

b) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$

c) $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

d) $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

Question 16. It is given that $z = -6 + 6\sqrt{3}i$, $|w| = 4$ and $\arg w = -\frac{5\pi}{12}$. Find each of the following in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$:

a) zw

b) $\frac{z}{w}$

Question 17. Points A and B on an argand diagram are given by $z_1 = 4\mathrm{e}^{-\frac{\pi}{6}\mathrm{i}}$ and $z_2 = 4\mathrm{e}^{\frac{\pi}{6}\mathrm{i}}$ respectively. Point C is given by a third complex number z_3 ACB forms an isosceles triangle where $|z_3| = 4$.

a) Sketch z_1 and z_2 on the same Argand diagram.

b) Find the possible angle(s) $\angle ACB$

Question 18. Use $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ to show that

a) $\cos\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$

b) $\sin\theta = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})$

Question 19. Use $e^{i\theta} = \cos\theta + i\sin\theta$ to show that

a) $\sin^2\theta + \cos^2\theta = 1$

b) $2\sin\theta\cos\theta = \sin 2\theta$

c) $\cos^2\theta - \sin^2\theta = \cos 2\theta$

Question 20. By writing $\cos\theta + i\sin\theta$ as $e^{i\theta}$, show that

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

Question 21. Given that $z = a + b\mathrm{i}$ for real numbers a and b, show that $\mathrm{e}^{z^*} = (\mathrm{e}^z)^*$

Question 22. Show that $\mathrm{e}^{2\mathrm{i}\theta} - 1 = 2\mathrm{i}\mathrm{e}^{\mathrm{i}\theta}\sin\theta$

Question 23. Express $\left(1-\frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}\right)\left(1-\frac{1}{4}\mathrm{e}^{-\mathrm{i}\theta}\right)$ in terms $\cos\theta$

Question 24. Express $(1-a\mathrm{e}^{\mathrm{i}\theta})(1-a\mathrm{e}^{-\mathrm{i}\theta})$ in terms of a and $\cos\theta$ where a is a real constant.

Question 25. Given that $z = 2e^{\frac{\pi}{6}i}$ and $w = 2e^{\frac{\pi}{3}i}$ show that

$$z + w = (\sqrt{2} + \sqrt{6})e^{\frac{\pi}{4}i}$$

Complex Numbers: De Moivre's Theorem - Multiple Angles

Questions 1–23

Question 1. By first expanding the brackets using a binomial expansion, find each of the following in its simplest form.

a) $(1+\mathrm{i})^3$
b) $(1+\mathrm{i})^4$
c) $(1+\mathrm{i})^5$
d) $(1+\mathrm{i})^6$

Question 2. By first expanding the brackets using a binomial expansion, find each of the following in its simplest form in terms of a and b where a and b are positive real constants.

a) $(a+b\mathrm{i})^3$
b) $(a+b\mathrm{i})^4$
c) $(a+b\mathrm{i})^5$

Question 3.

a) Find the expansion of $(\cos\theta + \mathrm{i}\sin\theta)^2$

b) Hence show that $\cos 2\theta = \cos^2\theta - \sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$

Question 4.

a) Find the expansion of $(\cos\theta + \mathrm{i}\sin\theta)^3$

b) Hence show that

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta \text{ and } \sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

c) Hence find $\tan 3\theta$ in terms of powers of $\tan\theta$ only.

Question 5. Given $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

a) Solve $4x^3 - 3x = 1$

b) Find the solutions to $\cos 3\theta = 1$ for $0 \leq \theta \leq \pi$

c) Hence, by considering $x = \cos\theta$, verify all the solutions to $4x^3 - 3x = 1$ in the interval $-1 \leq x \leq 1$

Question 6.

a) Use De Moivre's theorem to show that

$$\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$$

b) By using $\cos 4\theta = -1$, find the exact values to the equation $x^4 - x^2 = -\frac{1}{4}$

Prerequisite knowledge for Q7: Reversing the Chain Rule

Question 7.

a) Show that

$$\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$$

b) Hence show that

$$\int_0^{\frac{3\pi}{4}} (\cos^3\theta\sin\theta - \cos\theta\sin^3\theta)\,\mathrm{d}\theta = \frac{1}{8}$$

Question 8. Use De Moivre's theorem to show that

$$\cot 3\theta = \frac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta - 1}$$

Question 9.

a) Show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

b) Show that $\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$

c) Find $(1 + \tan^2\theta)^2$

d) Hence, show that

$$\tan 5\theta = \frac{\tan^5\theta - 10\tan^3\theta + 5\tan\theta}{1 - 10\tan^2\theta + 5\tan^4\theta}$$

e) Find the five distinct real solutions to

$$1 - 10x^2 + 5x^4 = x^5 - 10x^3 + 5x$$

giving your answers to three decimal places.

Question 10.

a) Find $(1 - \cos^2\theta)^3$

b) Show that

$$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$

c) Hence, find the six solutions to

$$64x^6 - 96x^4 + 36x^2 - 3 = 0$$

giving your answers to four decimal places.

Question 11. Show that

$$\sin 6\theta = a\cos^5\theta\sin\theta - b\cos^3\theta\sin^3\theta + c\cos\theta\sin^5\theta$$

where a, b and c are integers to be found.

Question 12.

a) Find $\frac{\sin 4\theta}{\sin\theta}$ in terms of $\cos\theta$

b) Hence solve the equation

$$\sin 4\theta = \sin\theta$$

for $0 < \theta < \pi$ giving your answers to three significant figures.

Question 13.

a) Find $\frac{\cos 5\theta}{\cos \theta}$ in terms of $\cos \theta$

b) Hence solve the equation

$$\cos 5\theta = -\cos \theta$$

for $0 < \theta \leq \pi$ giving your answers in exact form.

Question 14. It is given that $z = 1 + 3\mathrm{i}$

a) Find z^3 in the form $a + b\mathrm{i}$ where a and b are real numbers.

b) Let $\theta = \arg z$. Find $\arg z$ giving your answer in exact form.

c) Use De Moivre's theorem to show that

$$\cos 3\theta = 4\cos^3 \theta - 3\cos \theta \text{ and } \sin 3\theta = 3\sin \theta - 4\sin^3 \theta$$

d) Show that

$$\tan 3\theta = \frac{3\tan \theta - \tan^3 \theta}{1 - 3\tan^2 \theta}$$

e) Use your results to verify that the value of $\tan \theta$ obtained from z is consistent with the identity for $\tan 3\theta$

Prerequisite Knowledge for Q15-Q18: Hyperbolic Functions and Euler's Formula

Question 15.

a) Expand $(\cosh x + \sinh x)^3$

b) Show that

$$(\cosh x + \sinh x)^3 = \cosh 3x + \sinh 3x$$

and $(\cosh x - \sinh x)^3 = \cosh 3x - \sinh 3x$

c) Hence, find $\cosh 3x$ and $\sinh 3x$ in terms of $\cosh x$ and $\sinh x$

d) Given that $\cosh x = 2$ and $\sinh x = \sqrt{3}$, determine the exact value of $\cosh 3x$ and $\sinh 3x$

Question 16.

a) By using Euler's formula show that $\cosh(\mathrm{i}z) = \cos z$

b) Hence, find an exact value for $\cosh\left(\frac{3\pi}{4}\mathrm{i}\right)$

Question 17.

a) By using Euler's formula show that $\sinh(\mathrm{i}z) = \mathrm{i}\sin z$

b) Hence, find an exact value for $\sinh\left(\frac{5\pi}{6}\mathrm{i}\right)$

Question 18. Given $\sinh(\mathrm{i}z) = \mathrm{i}\sin(z)$, show that

$$\sinh\left(\frac{3\pi}{7}\mathrm{i}\right) = 3\mathrm{i}\cos^2\left(\frac{\pi}{7}\right)\sin\frac{\pi}{7} - \mathrm{i}\sin^3\left(\frac{\pi}{7}\right)$$

Question 19. Use De Moivre's theorem to find $\cos 3\theta$ and hence show that

$$x = \cos\left(\frac{4\pi}{9}\right)$$

satisfies $8x^3 - 6x + 1 = 0$

Question 20.

a) Show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

b) By considering the solutions of $\cos(5\theta) = 0$ for $0 \leq \theta \leq \pi$, show that

$$\cos^2\left(\frac{\pi}{10}\right) = \frac{5+\sqrt{5}}{8}$$

and

$$\cos^2\left(\frac{3\pi}{10}\right) = \frac{5-\sqrt{5}}{8}$$

Question 21.

a) Show that

$$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$

b) By considering the solutions of $\cos 6\theta = 0$ for $0 \leq \theta \leq \pi$, show that

$$\cos^2\left(\frac{\pi}{12}\right) = \frac{2+\sqrt{3}}{4}$$

and

$$\cos^2\left(\frac{5\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

Question 22.

a) Use De Moivre's theorem to find $\cos 5\theta$

b) By considering $\cos 5\theta = 1$, show that x satisfies the equation

$$(x-1)(4x^2+2x-1)^2 = 0$$

c) Hence, show that

$$\cos 72^\circ = \frac{-1+\sqrt{5}}{4}$$

and

$$\cos 144^\circ = \frac{-1-\sqrt{5}}{4}$$

Prerequisite Knowledge for Q23: Geometric Series

Question 23.

a) Find $\cos n\theta$ in terms of $\sin \frac{n\theta}{2}$

b) Find $\sin n\theta$ in terms of $\sin \frac{n\theta}{2}$ and $\cos \frac{n\theta}{2}$

c) Show that

$$\sum_{r=1}^{n} \cos(r\theta) = \frac{\cos\left(\frac{(n+1)\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

d) Show that

$$\sum_{r=1}^{n} \sin(r\theta) = \frac{\sin\left(\frac{(n+1)\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

Complex Numbers: De Moivre's Theorem - Powers

Questions 1–33

Question 1. Given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Find z^{-1}

b) Hence show that $2\cos\theta = z + z^{-1}$ and $2\mathrm{i}\sin\theta = z - z^{-1}$

Question 2.

a) Find z^{-n}

b) Show that $\cos n\theta = \frac{z^n + z^{-n}}{2}$

c) Show that $\sin n\theta = \frac{z^n - z^{-n}}{2\mathrm{i}}$

Question 3. Find the expansion of each of the following

a) $\left(z+\frac{1}{z}\right)^2$

b) $\left(z+\frac{1}{z}\right)^3$

c) $\left(z+\frac{1}{z}\right)^4$

d) $\left(z+\frac{1}{z}\right)^5$

e) $\left(z+\frac{1}{z}\right)^6$

Question 4. Find the expansion of each of the following

a) $\left(z-\frac{1}{z}\right)^2$

b) $\left(z-\frac{1}{z}\right)^3$

c) $\left(z-\frac{1}{z}\right)^4$

d) $\left(z-\frac{1}{z}\right)^5$

e) $\left(z-\frac{1}{z}\right)^6$

Prerequisite Knowledge for Q5-Q7: Exponential Form

Question 5. Use Euler's formula to write each of the following in terms of $e^{i\theta}$ and $e^{-i\theta}$

a) $\cos\theta$

b) $\sin\theta$

Question 6. Find the expansion of each of the following:

a) $(e^{i\theta} + e^{-i\theta})^2$

b) $(e^{i\theta} + e^{-i\theta})^3$

c) $(e^{i\theta} + e^{-i\theta})^4$

Question 7. Find the expansion of each of the following:

a) $\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})\right]^2$

b) $\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})\right]^3$

c) $\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})\right]^4$

Question 8. Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z + \frac{1}{z}\right)^2$ to show that $\cos^2\theta = \frac{1}{2}\cos 2\theta + \frac{1}{2}$

Question 9. Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^2$ to show that $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$

Prerequisite knowledge for Q10: Exponential Form and Integration of Standard Functions

Question 10.

a) By writing $\cos\theta$ in terms of $\mathrm{e}^{\mathrm{i}\theta}$ and $\mathrm{e}^{-\mathrm{i}\theta}$ show that $\cos^3\theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta$

b) Hence show that

$$\int_0^{\frac{\pi}{3}} \cos^3\theta \,\mathrm{d}\theta = \frac{p}{q}\sqrt{3}$$

where p and q are constants to be found.

Question 11.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^3$ to show that $\sin^3\theta = \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta$

b) Hence solve $\frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta = \frac{1}{8}$ in the interval $0 \leq \theta \leq 2\pi$

Question 12.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^4$ to show that

$$\sin^4\theta = \frac{1}{8}(a\cos 4\theta + b\cos 2\theta + c)$$

where a, b and c are constants to be found.

b) Hence show that

$$\sin\frac{\pi}{8} = \left(\frac{3 - 2\sqrt{2}}{8}\right)^{\frac{1}{4}}$$

Prerequisite knowledge for Q13: Reversing the Chain Rule

Question 13.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z + \frac{1}{z}\right)^4$ to find $\cos^4\theta$ in terms of cosines of multiple angles.

b) Hence show that

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 8\cos^4\theta \,\mathrm{d}\theta = 3\pi$$

Prerequisite Knowledge for Q14-Q15: Exponential Form

Question 14. Will is proving that

$$32\sin^4\theta \equiv 4\cos 4\theta - 16\cos 2\theta + 12$$

His workings are shown below:

$$\begin{aligned}
16\sin^4\theta &\equiv (2\sin\theta)^4 \\
&\equiv (\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})^4 \\
&\equiv 1(\mathrm{e}^{\mathrm{i}\theta})^4 + 4(\mathrm{e}^{\mathrm{i}\theta})^3(\mathrm{e}^{-\mathrm{i}\theta}) + 6(\mathrm{e}^{\mathrm{i}\theta})^2(\mathrm{e}^{-\mathrm{i}\theta})^2 \\
&\quad + 4(\mathrm{e}^{\mathrm{i}\theta})(\mathrm{e}^{-\mathrm{i}\theta})^3 + (\mathrm{e}^{-\mathrm{i}\theta})^4 \\
&\equiv \mathrm{e}^{4\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta} + 4\mathrm{e}^{2\mathrm{i}\theta} + 4\mathrm{e}^{-\mathrm{i}2\theta} + 6 \\
&\equiv 2\sin 4\theta + 4\sin 2\theta + 6
\end{aligned}$$

So that

$$32\sin^4 x \equiv 4\sin 4\theta + 8\sin 2\theta + 12$$

a) Find the error in his working.

b) Fully correct his working.

c) Use a similar method to show that

$$32\cos^4\theta \equiv 4\cos 4\theta + 16\cos 2\theta + 12$$

Question 15.

a) By writing $\cos\theta$ in terms of $e^{i\theta}$ and $e^{-i\theta}$ find $\cos^5\theta$ in terms of cosines of multiple angles.

b) Hence solve

$$2\cos 5\theta + 10\cos 3\theta + 20\cos\theta - 1 = 0$$

in the interval $0 \leq \theta \leq 2\pi$ giving your answers in exact form.

Prerequisite knowledge for Q16: Reversing the Chain Rule

Question 16.

a) Given $z = \cos\theta + i\sin\theta$, use $\left(z - \frac{1}{z}\right)^5$ to find $\sin^5\theta$ in terms of cosines of multiple angles.

b) Hence show that

$$\int_0^{\frac{\pi}{2}} 16\sin^5\theta \, d\theta = \frac{128}{15}$$

Question 17.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z + \frac{1}{z}\right)^6$ to find $\cos^6\theta$ in terms of cosines of multiple angles.

b) Hence show that

$$\cos\frac{\pi}{12} = \frac{1}{2}(26 + 15\sqrt{3})^{\frac{1}{6}}$$

Prerequisite knowledge for Q18: Exponential Form

Question 18.

a) By writing $\sin\theta$ in terms of $\mathrm{e}^{\mathrm{i}\theta}$ and $\mathrm{e}^{-\mathrm{i}\theta}$ show that

$$32\sin^6\theta = -\cos 6\theta + a\cos 4\theta - b\cos 2\theta + c$$

where a, b and c are positive constants to find.

b) Hence show that

$$\sin\frac{\pi}{12} = \frac{1}{2}(26 - 15\sqrt{3})^{\frac{1}{6}}$$

Question 19. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$. By expanding $\left(z^3 + \frac{1}{z^3}\right)^2$, show that

$$2\cos^2 3\theta = 1 + \cos 6\theta$$

Question 20. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$. By expanding $\left(z^4 - \frac{1}{z^4}\right)^3$, show that

$$4\sin^3 4\theta = 3\sin 4\theta - \sin 12\theta$$

Question 21. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$. By expanding $\left(z^3 - \frac{1}{z^3}\right)^5$, to find an expression for $\sin^5 3\theta$ in terms of sines of multiple angles.

Question 22. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Find $z^n + \frac{1}{z^n}$ in trigonometric form.

b) Find $z^n - \frac{1}{z^n}$ in trigonometric form.

c) Hence, using $\left(z + \frac{1}{z}\right)^2\left(z - \frac{1}{z}\right)^2$ show that

$$\sin^2\theta\cos^2\theta = \frac{1}{8}(1 - \cos 4\theta)$$

Question 23. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Find $z^n + \frac{1}{z^n}$ in trigonometric form.

b) Find $z^n - \frac{1}{z^n}$ in trigonometric form.

c) Hence, using $\left(z + \frac{1}{z}\right)^3\left(z - \frac{1}{z}\right)^3$ show that

$$\sin^3\theta\cos^3\theta = \frac{1}{32}(3\sin 2\theta - \sin 6\theta)$$

Question 24. Without using a trigonometric identity, show that

$$\cos^3\theta\sin^4\theta = \frac{1}{64}(a\cos 7\theta + b\cos 5\theta + c\cos 3\theta + d\cos\theta)$$

where a, b, c and d are integers to be found.

Prerequisite knowledge for Q25-Q26: Exponential Form

Question 25.

a) Show that

$$\mathrm{Re}\left(\mathrm{e}^{3\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^3\right) = k\cos 3\theta\cos^3\theta$$

where k is an integer to be found.

b) Hence show that

$$\cos\frac{\pi}{12} = \sqrt[3]{\frac{p\sqrt{2}+q\sqrt{6}}{16}}$$

where p and q are integers to be found.

Question 26.

a) Show that

$$\mathrm{Im}\left(\mathrm{e}^{4\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})^4\right) = k\sin 4\theta\sin^4\theta$$

where k is an integer to be found.

b) Hence show that

$$\sin\frac{\pi}{8} = \frac{1}{2}(p-q\sqrt{2})^{\frac{1}{4}}$$

where p and q are integers to be found.

Prerequisite knowledge for Q27-Q29: Integration by Substitution and/or Reversing the Chain Rule

Question 27. Find $\int \cos^4\theta \sin\theta \, \mathrm{d}\theta$:

a) By using standard integration methods.

b) By first writing $\cos^4\theta \sin\theta$ in terms of sines of multiple angles.

Question 28.

a) Show that

$$16\sin^3\theta\cos^2\theta = a\sin\theta + b\sin 3\theta + c\sin 5\theta$$

where a, b and c are constants to be found.

b) Hence, by using $x = \sin\theta$ show that

$$\int_0^1 16x^3\sqrt{1-x^2}\,\mathrm{d}x = \frac{32}{15}$$

Question 29.

a) Show that

$$8\sin^2\theta\cos^2\theta = a - b\cos 4\theta$$

where a and b are constants to be found.

b) Hence, by using a suitable substitution show that

$$\int_0^2 4x^2\sqrt{4-x^2}\,\mathrm{d}x = 4\pi$$

Prerequisite knowledge for Q30: Reduction Formulae for Integration

Question 30.

a) Given $z = \cos\theta + \mathrm{i}\sin\theta$, use $\left(z - \frac{1}{z}\right)^6$ to write $\sin^6\theta$ in terms of cosines of multiple angles.

b) Hence show that

$$\int_0^{\frac{\pi}{4}} \sin^6\theta\,\mathrm{d}\theta = \frac{p\pi - q}{r}$$

where p, q and r are constants to be found.

c) Verify your result in part b) by using the formula

$$I_n = \left(\frac{n-1}{n}\right) I_{n-2} - \frac{1}{n2^{\frac{n}{2}}}$$

where $I_n = \int_0^{\frac{\pi}{4}} \sin^n\theta\,\mathrm{d}\theta$

Question 31. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Show that $z^n + \frac{1}{z^n} = 2\cos n\theta$

b) Show that $(2x-1)$ is a factor of $8x^3 - 12x^2 + 6x - 1$

c) Use $\left(z + \frac{1}{z}\right)^3$ to show that

$$8\cos^3\theta = 6\cos\theta + 2\cos 3\theta$$

d) Hence, by considering $\left(z + \frac{1}{z}\right)^3$, $\left(z + \frac{1}{z}\right)^2$ and $z + \frac{1}{z}$ solve

$$z^6 - 3z^5 + 6z^4 - 7z^3 + 6z^2 - 3z + 1 = 0$$

giving your answers in the form $a + b\mathrm{i}$ where a and b are constants to be found.

Question 32. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Show that $z^n + \frac{1}{z^n} = 2\cos n\theta$

b) Use $\left(z + \frac{1}{z}\right)^3$ to show that

$$\cos^3\theta = \frac{1}{4}(3\cos\theta + \cos 3\theta)$$

c) It is given that

$$f(z) = z^6 + 3z^4 + z^3 + 3z^2 + 1$$

Considering $f(z) = 0$, write $f(z)$ as a product of a quadratic and a quartic factor.

Question 33. It is given that $z = \cos\theta + \mathrm{i}\sin\theta$

a) Use $\left(z + \frac{1}{z}\right)^5$ to show that

$$32\cos^5\theta = 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta$$

b) Hence write

$$z^{10} + 5z^8 + 10z^6 - z^5 + 10z^4 + 5z^2 + 1$$

as a product of two polynomials.

Complex Numbers: Roots of Unity

Questions 1–30

Question 1. Convert the following points into the form $a + b\mathrm{i}$ where a and b are real constants:

a) $\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$
b) $\mathrm{e}^{\frac{3\pi}{4}\mathrm{i}}$
c) $\mathrm{e}^{\pi\mathrm{i}}$
d) $\mathrm{e}^{\frac{3\pi}{2}\mathrm{i}}$
e) $\mathrm{e}^{\frac{7\pi}{6}\mathrm{i}}$
f) $\mathrm{e}^{2\pi\mathrm{i}}$

Question 2.

a) Find the three roots of the equation $z^3 = 1$ giving your answers in cartesian form.
b) Plot the three roots on an Argand diagram.
c) State the name of the polygon formed by the points representing the roots.
d) Find the sum of the roots.
e) Find the area of the shape formed by the roots giving your answer in exact form.

Question 3.

a) Find the five roots of the equation $z^5 = 1$ giving your answers in exponential form.
b) Plot the five roots on an Argand diagram.
c) State the name of the polygon formed by the points representing the roots.
d) Find the sum of the roots.
e) Find the area of the shape formed by the roots giving your answer to three significant figures.

Question 4.

a) Find the six roots of the equation $z^6 = 1$ giving your answers in cartesian form.
b) Plot the six roots on an Argand diagram.
c) Find the sum of the roots.
d) Find the area of the shape formed by the roots giving your answer in exact form.

Question 5.

a) Find all the roots of the equation $z^7 = 1$ giving your answers in exponential form.

b) Plot the seven roots on an Argand diagram.

c) Find the sum of the roots.

d) Find the area of the shape formed by the roots giving your answer to three significant figures.

Question 6. Given that $z^n = 1$ for positive integer n

a) Find the area of the shape formed by the roots giving your answer in terms of n

b) Find the area of the shaped formed by the roots as $n \to \infty$

Question 7. Prove that the distance between any two consecutive n^{th} roots of unity is $2\sin\left(\frac{\pi}{n}\right)$

Question 8. An equilateral triangle is centred at the origin with one vertex at $(2, 0)$

a) Find the coordinates of the other vertices.

b) Find the area of the triangle giving your answer in exact form.

Question 9. A regular hexagon is centred at the origin with one vertex at $(3, 0)$

a) Find the coordinates of the other vertices.

b) Find the area of the hexagon.

Question 10. Show that if z is an nth root of unity then its complex conjugate is also an nth root of unity.

Question 11. Prove that the sum of the n roots of unity is zero.

Question 12. Show that

a) $\omega = \cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}$ is a solution to $z^5 = 1$

b) $\omega = \mathrm{e}^{\frac{2\pi}{7}\mathrm{i}}$ is a root of the equation $z^7 = 1$

Question 13.

a) Find the five roots of the equation $z^5 - 1 = 0$ giving your answer in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$

b) Hence show that

$$\cos\left(\frac{2\pi}{5}\right) + \cos\left(\frac{4\pi}{5}\right) = -\frac{1}{2}$$

Question 14.

a) Find the seven roots of the equation $z^7 - 1 = 0$ giving your answer in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \theta$

b) Hence show that

$$\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right) = -\frac{1}{2}$$

Question 15. By considering the fifth roots of unity, prove

$$\cos\left(\frac{\pi}{5}\right) + \cos\left(\frac{3\pi}{5}\right) = \frac{1}{2}$$

Question 16. Derive a cubic polynomial with integer coefficients, expressed in expanded form, whose roots are:

a) the three cube roots of unity.

b) the midpoints on the Argand diagram of the three cube roots of unity.

Question 17. It is given that

$$f(z) = z^5 + z^4 + z^3 + z^2 + z + 1$$

a) Find all of the solutions to $f(z) = 0$
b) Hence fully factorise $f(z)$

Question 18. $f(z) = z^7 + z^6 + z^5 + z^4 + z^3 + z^2 + z + 1$
a) Find all of the solutions to $f(z) = 0$
b) Hence fully factorise $f(z)$

Working with ω

Question 19. If ω is a complex cube root of unity and $\omega \neq 1$, without directly using the roots of unity simplify:

a) ω^3

b) ω^5

c) $\omega^8 - \omega^5 + \omega^6$

d) $\frac{1}{\omega}$

Question 20. Given ω is an nth root of unity, by considering the roots in their exponential form,

a) show that the complex conjugate of ω is ω^{n-1}

b) Hence, find the complex conjugate of ω^k for integer $k < n$

Question 21. Let $\omega = \mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$ where ω is a root of unity, find

a) $\mathrm{Re}(\omega + \omega^2 + \omega^3 + \omega^4 + \omega^5)$

b) $\mathrm{Im}(\omega^2 + \omega^4)$

Question 22. Let $\omega = \mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$ where ω is a root of unity, find

a) $\mathrm{Re}(\omega^2 + \omega)$

b) $\mathrm{Im}(1 + \omega^4)$

Question 23. Given that ω is a complex 4th root of unity such that $\omega \neq 1$, show that

$$(1-\omega)(1-\omega^2)(1-\omega^3) = 4$$

Question 24.

a) Simplify $1 + x + x^2 + \cdots + x^{n-1}$

b) Given ω is a principal n^{th} root of unity so that $z^n - 1 = 0$, show that

$$\prod_{k=1}^{n-1}(1-\omega^k) = n$$

Question 25.

a) Prove that $|z_1 + z_2|^2 = (z_1 + z_2)(z_1^* + z_2^*)$

b) Given that ω is a cube root of unity, show that

$$|\omega - \omega^2|^2 = 2 - (\omega + \omega^*)$$

Question 26. If ω is a complex cube root of unity and $\omega \neq 1$, without directly using the roots of unity find

a) $(1 - \omega)(1 - \omega^2)$

b) $(1 - \omega)(1 - \omega^2)(1 - \omega^3)$

c) $(1 + \omega)(1 + \omega^2)$

d) $(1 + \omega)(1 + \omega^2)(1 + \omega^3)$

Question 27. If ω is a complex cube root of unity such that $\omega \neq 1$, and a and b are real constants, find each of the following in terms of a and b:

a) $(a - b\omega)(a - b\omega^2)$
b) $(a - b\omega)(a - b\omega^2)(a - b\omega^3)$
c) $(a + b\omega)(a + b\omega^2)$
d) $(a + b\omega)(a + b\omega^2)(a + b\omega^3)$

Question 28. If ω is a complex 4th root of unity and $\omega \neq 1$, without directly using the roots of unity find

a) $\omega(1 + \omega + \omega^2)$
b) $\omega(1 + \omega)(1 + \omega^2)$
c) $\left(\omega + \frac{1}{\omega}\right)^2 + 2\left(\omega + \frac{1}{\omega}\right)$

Question 29. If ω is a complex 3rd root of unity and $\omega \neq 1$,

a) find $\omega + \frac{1}{\omega} + 1$

b) Hence, using your result from part a), prove that $\cos \frac{4\pi}{3} = -\frac{1}{2}$

Question 30. If ω is a complex 5th root of unity and $\omega \neq 1$,

a) find $\left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega + \frac{1}{\omega}\right)$

b) Hence show that

$$\cos \frac{4\pi}{5} = \frac{p + q\sqrt{5}}{r}$$

where p, q and r are constants to be found.

Complex Numbers: The nth Roots of a Complex Number

Questions 1–25

Question 1. Find all the solutions to each of the following equations, giving your answer in the form $a + bi$ where a and b are constants:

a) $z^3 - 8i = 0$
b) $z^3 + 27 = 0$
c) $z^2 = 1 + \sqrt{3}i$
d) $z^4 + 16 = 0$

Question 2. Find all the solutions to each of the following equations, giving your answers in the exact form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$:

a) $z^5 - 32i = 0$
b) $z^3 = 1 + \sqrt{3}i$
c) $z^4 = -2\sqrt{2} + 2\sqrt{2}i$
d) $z^8 = 8 + 8\sqrt{3}i$

Question 3. Find all the solutions to each of the following equations, giving your answer in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$:

a) $z^7 - 128i = 0$

b) $z^6 = -32 - 32\sqrt{3}i$

c) $2z^4 - 9 = 9\sqrt{3}i$

Question 4. Find the cube roots of $-2 + 2i$ giving your answers in the form $a + bi$ where a and b are real constants rounded to three significant figures.

Question 5. Evaluate each of the following, giving your answers in the form $a+bi$ where a and b are real constants rounded to three significant figures:

a) $\sqrt{-2+2i}$
b) $\sqrt{-2+4i}$
c) $\sqrt[4]{4-4i}$
d) $\sqrt[3]{4+4i}$

Question 6. Find the two roots of the equation $(z-i)^2 = -4+12i$ giving your answers in the form $a+bi$ where a and b are real constants rounded to three significant figures.

Question 7. Find all the solutions of $(z-2)^3 = 8i$ giving your answers in the form $a+bi$ where a and b are exact real constants.

Question 8. It is given for real constants a and b that a root of $z^3 = a+bi$ is $-3\sqrt{3}+3i$.

a) Find the values of a and b.

b) Find all the roots of $z^3 = a+bi$ giving your answer in the form $p+qi$ where p and q are real constants.

c) Explain why the complex roots do not come in conjugate pairs in this case.

Question 9. It is given for real constants a and b that a root of $z^4 = a + bi$ is $1 + i$.

a) Find the values of a and b.

b) Find all the roots of $z^4 = a + bi$ giving your answer in the form $p + qi$ where p and q are real constants.

c) Explain why the complex roots do come in conjugate pairs in this case.

Question 10. It is given that $z^3 = 5 + 5\sqrt{3}i$.

a) Find the solutions to the equation giving your answers in the form $10^k e^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

b) Show that these roots satisfy the equation $w^9 + 1000 = 0$.

Question 11. Solve $z^8 = (2+2i)^4$ giving your answers in the form $2^k e^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

Question 12.

a) Write $1+i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Write $\sqrt{3}-i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

c) Determine the four roots of the equation $z^4 = (1+i)^2(\sqrt{3}-i)^2$ giving your answers in the form $2^k e^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

Question 13. It is given that $z = 5\sqrt{3} + 5i$.

a) Find z in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

b) Determine the values of w such that $w^4 = z^5$ giving your answers in the form $10^{5/4}(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

Question 14. Solve $z^{5/3} = -1 - \sqrt{3}i$ giving your answers in the form $2^k \mathrm{e}^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

Question 15. Solve $z^{4/3} = \sqrt{2} + \sqrt{2}i$ giving your answers in the form $2^k e^{i\theta}$ where $-\pi < \theta \leq \pi$ and k is a constant to be found.

Question 16. It is given that $z_1 = 1 + \sqrt{3}i$ and $z_2 = \sqrt{3} - i$.

a) Find z_1 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

b) Find z_2 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

c) Find all the values of w such that $w^6 = z_1^3 z_2^2$ giving your answers in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

Question 17.

a) Find the expansion of $(a-1)^3$.

b) Hence find the solutions to $z^3 - 3z^2 + 3z - 9 = 0$ giving your answers in the form $a + bi$ where a and b are real constants.

Question 18. Find all the six roots of $z^6 - 6z^5 + 15z^4 - 20z^3 + 15z^2 - 6z + 2 = 0$ giving your answers in the form $a + bi$ where a and b are real constants.

Question 19. It is given that $z = 1 + i$ and n is the smallest positive whole number such that z^n is a positive whole number.

a) Determine the value of n.
b) Find the value of z^n.
c) Write down the value of all roots of z^n for your value of n in the form $a + bi$ where a and b are real constants.

Question 20. It is given that $z = -7 + 7\sqrt{3}i$ and n is the smallest positive whole number such that z^n is a positive whole number.

a) Determine the value of n.
b) Find the value of z^n.
c) Write down all the roots of z^n for your value of n in the form $a + bi$ where a and b are real constants.

Question 21. It is given that $z = \frac{2\sqrt{3}+2i}{\sqrt{3}-i}$ and n is the smallest positive whole number such that z^n is a positive whole number.

a) Determine the value of n.

b) Find the value of z^n.

c) Write down all the roots of z^n for your value of n in the form $a + bi$ where a and b are real constants.

Question 22. Find all the solutions of $16z^4 = (z - i)^4$ giving your answers in the form $a + bi$ where a and b are exact real constants.

Question 23. It is given that $z^3 - 1 = \sqrt{3}i$.

a) Determine the three roots of the equation giving your answers in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Hence, show that $\cos\left(\frac{\pi}{9}\right) + \cos\left(\frac{5\pi}{9}\right) + \cos\left(\frac{7\pi}{9}\right) = 0$.

Question 24. It is given that $z^3 - 6\sqrt{3} = 6i$.

a) Determine the three roots of the equation giving your answers in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Hence, show that $\sin\left(\frac{\pi}{18}\right) - \sin\left(\frac{11\pi}{18}\right) + \sin\left(\frac{13\pi}{18}\right) = 0$.

Question 25. It is given that $z_1 = 1 + \sqrt{3}i$ and $z_2 = 1 - \sqrt{3}i$.

a) Write z_1 in the form $re^{i\theta_1}$ where $-\pi < \theta_1 \leq \pi$ and $r > 0$.

b) Write z_2 in the form $re^{i\theta_2}$ where $-\pi < \theta_2 \leq \pi$ and $r > 0$.

c) Show that $z_1^6 + z_2^6 = 128$ and $z_1^6 - z_2^6 = 0$. Given ω is a cube root of unity satisfying $\omega^3 = 1$ and $\omega \neq 1$, **d)** Evaluate and simplify $z_1^6 + \omega z_2^6 + \omega^2 (z_1 z_2)^6$ giving your answer in the form $a + bi$ where a and b are real constants.

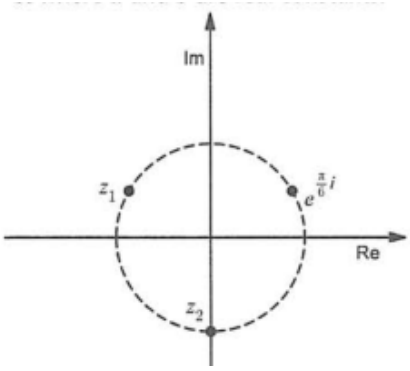
Complex Numbers: The Geometry of nth Roots

Questions 1–36

Question 1. For each given equation, sketch the solutions on an Argand diagram. In each case include a dashed line to represent the circle on which the roots lie.

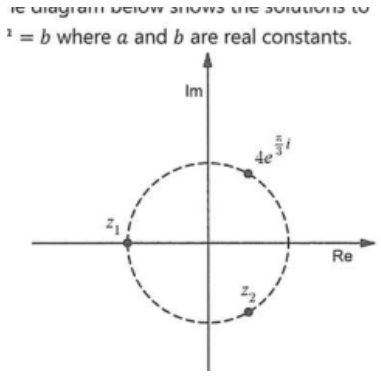
a) $z^3 = 1$
b) $z^3 = \mathrm{i}$
c) $z^5 = -\mathrm{i}$
d) $z^3 = -8$
e) $z^4 = -16\mathrm{i}$

Question 2. The diagram below shows the solutions to $z^a = bi$ where a and b are real constants.



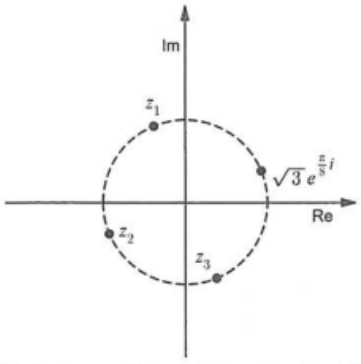
a) Write down all the solutions to $z^a = bi$ in the form $p + qi$ where p and q are real values to be found.
b) State the exact values of a and b.

Question 3. The diagram below shows the solutions to $z^a = b$ where a and b are real constants.



a) Write down all the solutions to $z^a = b$ in the form $p + qi$ where p and q are real values to be found.
b) State the exact values of a and b.

Question 4. The diagram below shows the solutions to $z^a = bi$ where a and b are real constants.



a) Write down all the solutions to $z^a = bi$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.
b) State the exact values of a and b.

Question 5. It is given that $z^3 = 8$.

a) Find the three roots of $z^3 = 8$ in the form $a + bi$ where a and b are real constants.

b) Plot the roots of $z^3 = 8$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

c) Hence, state the three roots of the equation $(z - i)^3 = 8$ in the form $a + bi$ where a and b are real constants.

d) Plot the roots of $(z - i)^3 = 8$ on an Argand diagram.

Question 6. It is given that $z^4 = 16i$.

a) Find the four roots of $z^4 = 16i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the roots of $z^4 = 16i$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

c) Hence, state the four roots of $(z - 2)^4 = 16i$ in the form $k + re^{i\theta}$ where $-\pi < \theta \leq \pi$, $r > 0$ and $k > 0$.

d) Plot the roots of $(z - 2)^4 = 16i$ on an Argand diagram.

Question 7. It is given that $z^6 = -64i$.

a) Find the six roots of $z^6 = -64i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the roots of $z^6 = -64i$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

c) Hence, state the six roots of $(z + 1 - i)^6 = -64i$ giving your answer in the form $a + bi$ where a and b are given to two decimal places.

d) Plot the roots of $(z + 1 - i)^6 = -64i$ on an Argand diagram.

Question 8. Find the coordinates of the vertices of each of the following regular polygons which all have centres at the origin, giving your answers to three significant figures:

a) Triangle with one vertex at $(1, 3)$

b) Pentagon with one vertex at $(3, 2)$

c) Hexagon with one vertex at $(2, 2)$

Question 9. Find the area each of the following regular polygons which all have centres at the origin giving your answers in exact form:

a) Triangle with one vertex at $(2, 2)$
b) Square with one vertex at $(5, 1)$
c) Hexagon with one vertex at $(3, 4)$

Question 10. Find the coordinates of the vertices of a regular hexagon with a vertex at $(2, 2)$ and with centre at $(0, 3)$ giving your answers to two decimal places.

Question 11. Find the coordinates of the vertices of an equilateral triangle with a vertex at $(2, -1)$ and with centre at $(1, 4)$.

Question 12. An equilateral triangle is such that it is centred at the origin and has area $3\sqrt{3}$. One of the vertices lies on the x-axis. Determine the coordinates of the vertices of the triangle giving your answers in exact form.

Question 13. An equilateral triangle is such that it is centred at $(1, 1)$ and has area $12\sqrt{3}$. One of the vertices is directly above the centre. Determine the coordinates of the other vertices of the triangle giving your answers in exact form.

Question 14. An equilateral triangle ABC has centre at the origin and a vertex at $A(3, 4)$.

a) Find the coordinates of points B and C giving your answers to two decimal places.

b) Find the exact area of the triangle ABC.

A triangle is formed using the coordinates of the midpoints of AB, BC and CA.

c) Find the area of the triangle formed by these three midpoints giving your answer in exact form.

Question 15. It is given that $f(z) = z^3 + a$ where a, b and c are real constants. The roots of the equation $f(z) = 0$ are z_1, z_2 and z_3. When plotted on an Argand diagram, the points representing these roots form the vertices of a triangle. The area of the triangle is $27\sqrt{3}$ and $\arg(z_1) = \frac{\pi}{3}$. Determine the value of a.

Question 16. It is given that $f(z) = z^3 + az^2 + bz + c$ where a, b and c are complex constants. The roots of the equation $f(z) = 0$ are z_1, z_2 and z_3. When plotted on an Argand diagram, the points representing these roots form the vertices of a triangle centred around $1 + i$. It is known that the area of the triangle is $3\sqrt{3}$ and $\text{Im}(z_1) = 1$ and $\text{Re}(z_1) > 0$. Determine the values of a, b and c.

Question 17. It is given that $f(z) = z^3 + az^2 + bz + c$ where a, b and c are real integers. The roots of the equation $f(z) = 0$ are z_1, z_2 and z_3. When plotted on an Argand diagram, the points representing these roots form the vertices of a triangle centred around $3 + 0i$. It is known that the area of the triangle is $\frac{27}{4}\sqrt{3}$ and $\text{Im}(z_1) = 0$ and $\text{Re}(z_1) > 0$. Determine the values of a, b and c.

Question 18. It is given that $f(z) = (z - 1)^3$.

a) Find the solution to $f(z) = 27i$ giving your answers in the form $a + bi$.

It is given further that the solutions to $f(z) = 27i$ represent the points A, B and C on an Argand diagram, where A and B have positive imaginary parts and A having the smaller real part.

b) Show that the area of the quadrilateral $OABC$ where O is the origin is $\frac{9}{4} + \frac{9\sqrt{3}}{2}$.

Question 19.

a) Find the cube roots of $-5 + 7\sqrt{5}i$ giving your answer in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the cube roots on an Argand diagram.

c) State the angles that any lines of symmetry of the triangle formed by the cube roots in the Argand diagram make with the positive real axis.

Question 20. Find the least value of $\arg z$ in the interval $-\pi < \arg z \leq \pi$ when $z^5 = \frac{\sqrt{3}}{2}(1 + i)$.

Question 21. Find the least value of $\arg z$ in the interval $-\pi < \arg z \leq \pi$ when $z^6 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$.

Question 22. Given two roots of $z^6 = i$, z_1 and z_2, lie in the positive quadrant of the Argand diagram. For $w = \frac{1}{2}(z_1 + z_2)$:

a) Find $\arg(w)$.

b) Hence, find the smallest value of n such that w^n is real.

Question 23. Given two roots of $z^5 = \sqrt{3} + i$, z_1 and z_2, lie in the positive quadrant of the Argand diagram. For $w = \frac{1}{2}(z_1 + z_2)$:

a) Find $\arg(w)$.

b) Hence, find the smallest value of n such that w^n is real.

Question 24. It is given that $z^3 = \sqrt{6} + \sqrt{2}i$.

a) Find the three roots of $z^3 = \sqrt{6} + \sqrt{2}i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the roots of $z^3 = \sqrt{6} + \sqrt{2}i$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

The transformation represented by the matrix $M = \begin{pmatrix} 6 & 2 \\ 2 & 4 \end{pmatrix}$ acts on the points plotted on the Argand diagram.

c) Find the exact area of the shape formed by the transformed points.

Question 25. It is given that $z^3 = 12\sqrt{3} + 36i$.

a) Find the three roots of $z^3 = 12\sqrt{3} + 36i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the roots of $z^3 = 12\sqrt{3} + 36i$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

The transformation represented by the matrix $M = \begin{pmatrix} 6 & 2 \\ -2 & -2 \end{pmatrix}$ acts on the points plotted on the Argand diagram.

c) Find the exact area of the shape formed by the transformed points.

Question 26.

a) Find all the roots of the equation $z^6 = 27$ in the form $a + bi$ where a and b are real constants.

b) Find the area of the shape formed on the Argand diagram of the roots of $z^6 = 27$.

The transformation represented by the matrix $M = \begin{pmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{pmatrix}$ acts on the points plotted on the Argand diagram.

These new points are the solutions to the equation $f(w) = 0$ given where $f(w) = w^6 + k$ with k a real constant.

c) Find the value of k.

d) Find the area of the shape formed by the roots of $f(z) = 0$.

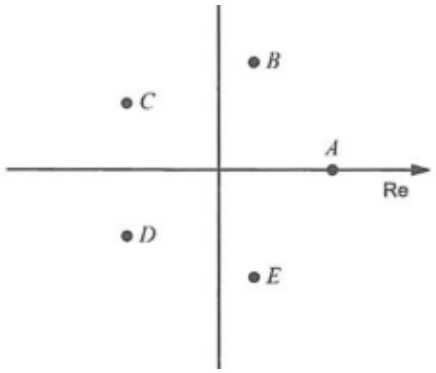
Question 27.

a) Show on an Argand diagram the points representing the three cube roots of unity.
b) Explain why, geometrically, the sum of the three cube roots of unity is zero.
c) Find the exact roots of the equation $z^3 - 1 = i$ in the form $r(\cos\theta + i\sin\theta)$ where $-\pi < \theta \leq \pi$ and $r > 0$.
d) Show that $\sin 15° + \sin 135° = \sin 105°$.

Question 28.

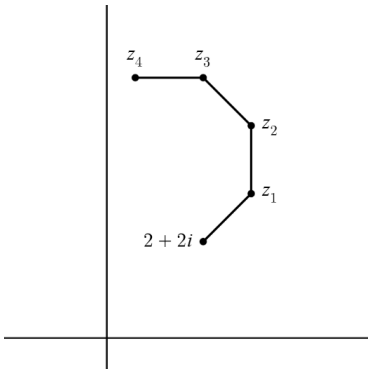
a) Find the exact roots of the equation $z^4 = 2\sqrt{3} + 2i$ in the form $r(\cos\theta + i\sin\theta)$ where $-\pi < \theta \leq \pi$ and $r > 0$.
b) Hence, show that $\cos 7.5° + \cos 82.5° + \cos 97.5° + \cos 172.5° = 0$.

Question 29. The roots of an equation are labelled A, B, C, D and E, and are shown in the diagram below. The distance between the origin and any of the points is 3 units.



a) State the equation of which the five roots are the solution.
b) Find the five roots of the equation giving your answers in the form $z = r(\cos\theta + i\sin\theta)$ for $-\pi < \theta \leq \pi$.
c) Show that a quadratic equation satisfied by B and E is $z^2 - 6z\cos\frac{2\pi}{5} + 9 = 0$.
d) Find a similar quadratic satisfied by the roots indicated by C and D.

Question 30. A particle is moving in the complex plane, with starting position $2 + 2i$. It initially faces the positive real direction. The particle rotates $\frac{\pi}{4}$ anticlockwise before moving in that direction a distance of $\sqrt{2}$. It repeats this a further three times.



Show that the displacement of the particle from its initial position is $2 - \sqrt{2} + (4 + \sqrt{2})i$.

Question 31.

a) Find the two roots of the equation $(z-1)^2 = 8 - 8\sqrt{3}i$ in the form $a+bi$ where a and b are real constants to be found.
b) Find $|z-1|^2$ for each root found in part a).
c) Find the sum of the squared distances from 1 for both roots found in part a).
d) Find the sum of the squared distances from the origin for both roots found in part a).

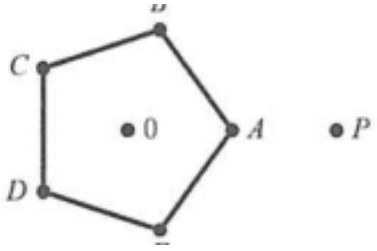
Question 32.

a) Find the roots of the equation $(z-3)^2 = -40 + 40\sqrt{3}i$ in the form $a+bi$ where a and b are real constants to be found.
b) Find $|z-3|^2$ for each root found in part a).
c) Find the sum of the squared distances from 3 for both roots found in part a).
d) Find the sum of the squared distances from the origin for both roots found in part a).

Question 33. Using the identity $zz^* = |z|^2$ show that if ω is any complex nth root of unity then $|\omega - 2|^2 = 5 - 2(\omega + \omega^*)$.

Question 34.

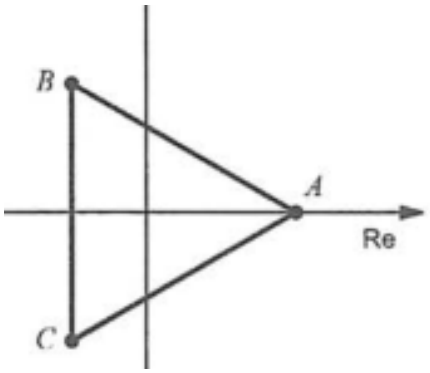
a) Using the identity $zz^* = |z|^2$ show that if ω is any complex nth root of unity then $|\omega - 3|^2 = 10 - 3(\omega + \omega^*)$.



Let $ABCDE$ be a regular pentagon whose vertices all lie on the unit circle with centre O and A at $(1, 0)$. Let the point P be the point $(3, 0)$. The squares of the distances from P to A, P to B etc. are denoted $(PA)^2 + (PB)^2 + (PC)^2 + (PD)^2 + (PE)^2$.

b) Find the value of $(PA)^2 + (PB)^2 + (PC)^2 + (PD)^2 + (PE)^2$.

Question 35. The Argand diagram below shows an equilateral triangle ABC centred about the origin where points A, B and C are complex numbers which are roots of $z^3 - a^3 = 0$.

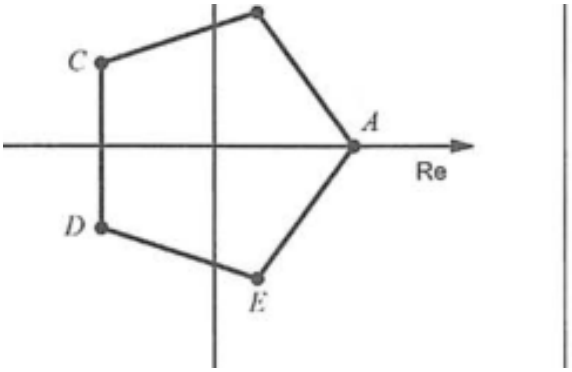


a) Find the points A, B and C in terms of a and in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$.
b) Show that the sum of the roots is zero.

A triangle is formed using the midpoints of AB, BC and CA.

c) Find a polynomial equation with roots of these midpoints. Give your equation in terms of a with integer coefficients.

Question 36. A regular pentagon $ABCDE$ centred about the origin where points A, B, C, D and E are complex numbers which are roots of $z^5 = a^5$.

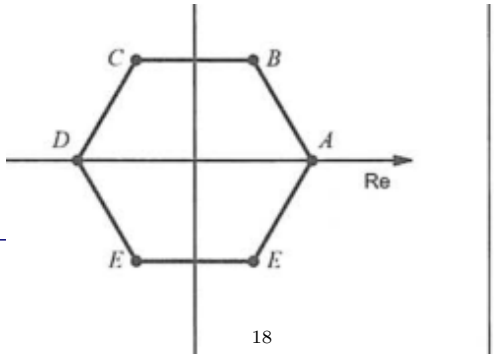


a) Find the points A, B, C, D and E in terms of a and in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$.
b) Show that the sum of the roots is zero.
c) Show that $e^{i\alpha} + e^{i\beta} = 2\cos\left(\frac{\alpha-\beta}{2}\right) e^{i(\alpha+\beta)/2}$.

A pentagon is formed using the midpoints of BC, CD, DE, EA and AB.

d) Show that a polynomial equation with roots of these midpoints is given by $z^5 - a^5\cos^5\left(\frac{\pi}{5}\right) = 0$.

A regular hexagon $ABCDEF$ is centred about the origin and points A, B, C, D, E and F are complex numbers which are roots of $z^6 = a^6$.



Complex Numbers: C + iS

Questions 1–43

Question 1. Identify whether each of the following series are a geometric series or a binomial series:

a) $1+\frac{1}{3}+\frac{1}{9}+\cdots$
b) $8-1.2+0.06-0.001$
c) $2-\frac{1}{2}+\frac{1}{8}-\cdots-\frac{1}{16384}$
d) $1+0.8+0.24+0.032+0.0016$

Question 2. Given that each of the following infinite series are convergent, find the sum of each giving your answer in the form $\frac{A}{B-e^{iC\theta}}$, where A, B and C are constants to be found.

a) $4+\frac{4e^{i\theta}}{3}+\frac{4e^{2i\theta}}{9}+\frac{4e^{3i\theta}}{27}+\cdots$
b) $1+\frac{e^{i\theta}}{4}+\frac{e^{2i\theta}}{16}+\frac{e^{3i\theta}}{64}+\cdots$
c) $2+\frac{2e^{2i\theta}}{5}+\frac{2e^{4i\theta}}{25}+\frac{2e^{6i\theta}}{125}+\cdots$
d) $5+\frac{e^{3i\theta}}{2}+\frac{e^{6i\theta}}{20}+\frac{e^{9i\theta}}{200}+\cdots$
e) $6+\frac{e^{-i\theta}}{2}+\frac{e^{-2i\theta}}{24}+\frac{e^{-3i\theta}}{288}+\cdots$
f) $2+\frac{e^{-4i\theta}}{3}+\frac{e^{-8i\theta}}{18}+\frac{e^{-12i\theta}}{108}+\cdots$

Question 3. Each of the following form a finite geometric series where the common ratio has a value not equal to 1. In each case, determine the sum:

a) $e^{i\theta} + e^{2i\theta} + e^{3i\theta} + \cdots + e^{12i\theta}$

b) $2 + e^{i\theta} + \frac{e^{2i\theta}}{2} + \cdots + \frac{e^{9i\theta}}{256}$

c) $1 - \frac{e^{i\theta}}{3} + \frac{e^{2i\theta}}{9} - \cdots + \frac{e^{8i\theta}}{6561}$

d) $16 - 4e^{2i\theta} + e^{4i\theta} - \cdots - \frac{e^{14i\theta}}{1024}$

Question 4. State the complex conjugate of each of the following:

a) $z = a + ib$

b) $z = r(\cos\theta + i\sin\theta)$

c) $z = re^{i\theta}$

Question 5. State the complex conjugate of each of the following:

a) $z = 5 + e^{i\theta}$
b) $z = 3 - e^{-i\theta}$
c) $z = 7 + e^{2i\theta}$
d) $z = 7 + 9e^{i\theta}$
e) $z = 2 - 3e^{i\theta}$
f) $z = 1 + e^{-i\theta}$
g) $z = 2 - e^{i\theta}$
h) $z = 5 - 5e^{-i\theta}$
i) $z = 6 - 8e^{-7i\theta}$
j) $z = 2 + 15e^{i\theta/2}$

Question 6. Evaluate and simplify each of the following:

a) $(1 + 11e^{-i\theta})(1 + 11e^{i\theta})$
b) $(4 + e^{3i\theta})(4 + e^{-3i\theta})$
c) $\left(2 + \frac{1}{2}e^{i\theta}\right)\left(2 + \frac{1}{2}e^{-i\theta}\right)$

Question 7. It is given that $z = e^{i\theta}$.

a) Write z in terms of sine and cosine.

b) Show that $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$.

Question 8. Write each of the following in the form $a + b\cos(p\theta)$, where a, b and p are constants to be found:

a) $10 + 6e^{i\theta} + 6e^{-i\theta}$

b) $5 - 2e^{4i\theta} - 2e^{-4i\theta}$

c) $(1 + e^{3i\theta})(1 + e^{-3i\theta})$

d) $(2 + e^{2i\theta})(2 + e^{-2i\theta})$

e) $\left(1 - \frac{1}{2}e^{i\theta/3}\right)\left(1 - \frac{1}{2}e^{-i\theta/3}\right)$

Question 9. Rationalise the denominator for each of the following, in each case using $2\cos\theta = e^{i\theta} + e^{-i\theta}$ to write the denominator in terms of $\cos\theta$:

a) $\frac{1}{2-e^{i\theta}}$

b) $\frac{3}{4+e^{2i\theta}}$

c) $\frac{5}{6-2e^{-i\theta}}$

d) $\frac{e^{i\theta}}{1-\frac{1}{3}e^{3i\theta}}$

e) $\frac{e^{i\theta}}{1+\frac{1}{8}e^{4i\theta}}$

Question 10. Write $C + iS$ as a sum of three exponential terms, given that $C = \cos\theta + \frac{1}{4}\cos 3\theta + \frac{1}{16}\cos 5\theta$ and $S = \sin\theta + \frac{1}{4}\sin 3\theta + \frac{1}{16}\sin 5\theta$.

Question 11. Write $C + iS$ as a sum of three exponential terms, given that $C = 1 - \frac{1}{2}\cos 2\theta + \frac{1}{4}\cos 4\theta$ and $S = -\frac{1}{2}\sin 2\theta + \frac{1}{4}\sin 4\theta$.

Question 12. Convergent infinite series C and S are defined by $C = \cos\theta - \frac{1}{3}\cos 4\theta + \frac{1}{9}\cos 7\theta - \cdots$ and $S = \sin\theta - \frac{1}{3}\sin 4\theta + \frac{1}{9}\sin 7\theta - \cdots$.

a) Expand and simplify $\left(1 + \frac{1}{3}e^{3i\theta}\right)\left(1 + \frac{1}{3}e^{-3i\theta}\right)$, writing your answer in the form $a + b\cos(3\theta)$, where a and b are constants to be found.

b) Find the sum of $C + iS$.

c) Show that $C = \frac{9\cos\theta + 3\cos 2\theta}{10 + 6\cos 3\theta}$.

d) Show that $S = \frac{9\sin\theta - 3\sin 2\theta}{10 + 6\cos 3\theta}$.

Question 13. Convergent infinite series C and S are defined by $C = \cos 3\theta + \frac{1}{2}\cos 5\theta + \frac{1}{4}\cos 7\theta + \cdots$ and $S = \sin 3\theta + \frac{1}{2}\sin 5\theta + \frac{1}{4}\sin 7\theta + \cdots$.

a) Show that $C = \frac{4\cos 3\theta - 2\cos\theta}{5 - 4\cos 2\theta}$.

b) Show that $S = \frac{4\sin 3\theta - 2\sin\theta}{5 - 4\cos 2\theta}$.

Question 14. Convergent infinite series C and S are defined by $C = 3 + \cos\theta + \frac{1}{3}\cos 2\theta + \cdots$ and $S = \sin\theta + \frac{1}{3}\sin 2\theta + \cdots$.

a) Show that $C = \frac{27 - 9\cos\theta}{10 - 6\cos\theta}$.

b) Show that $S = \frac{9\sin\theta}{10 - 6\cos\theta}$.

Question 15. Finite series C and S are defined by $C = 2 + \cos 2\theta + \frac{1}{2}\cos 4\theta + \cdots + \frac{1}{256}\cos 18\theta$ and $S = \sin 2\theta + \frac{1}{2}\sin 4\theta + \cdots + \frac{1}{256}\sin 18\theta$.

a) Show that $\left(1 - \frac{1}{2}e^{2i\theta}\right)\left(1 - \frac{1}{2}e^{-2i\theta}\right) = \frac{5}{4} - \cos 2\theta$.

b) Show that $C = \frac{2048 - 1024\cos 2\theta - 2\cos 20\theta + \cos 18\theta}{256(5 - 4\cos 2\theta)}$.

c) Show that $S = \frac{1024\sin 2\theta - 2\sin 20\theta + \sin 18\theta}{256(5 - 4\cos 2\theta)}$.

Question 16. Finite series C and S are defined by $C = \cos\theta + \cos 3\theta + \cos 5\theta + \cdots + \cos 45\theta$ and $S = \sin\theta + \sin 3\theta + \sin 5\theta + \cdots + \sin 45\theta$.

a) Show that $(1 - e^{2i\theta})(1 - e^{-2i\theta}) = 2 - 2\cos 2\theta$.

b) Show that $C = \frac{-\cos 47\theta + \cos 45\theta}{2 - 2\cos 2\theta}$.

c) Show that $S = \frac{2\sin\theta - \sin 47\theta + \sin 45\theta}{2 - 2\cos 2\theta}$.

Question 17.

a) Show that $\left(1-\frac{1}{9}e^{3i\theta}\right)\left(1-\frac{1}{9}e^{-3i\theta}\right)=\frac{1}{81}(82-18\cos 3\theta)$.

b) Find the sum of the convergent series $e^{3i\theta}+\frac{1}{9}e^{6i\theta}+\frac{1}{81}e^{9i\theta}+\cdots$, giving your answer in terms of $e^{i\theta}$.

c) Hence show that $C=\frac{81\cos 3\theta-9}{82-18\cos 3\theta}$.

d) Find a similar expression for $S=\sum_{n=1}^{\infty}\frac{1}{9^{n-1}}\sin(3n\theta)$.

Question 18. $C=\frac{1}{3}\cos\theta+\frac{1}{9}\cos 2\theta+\frac{1}{27}\cos 3\theta+\cdots+\frac{1}{3^n}\cos n\theta$ and $S=\frac{1}{3}\sin\theta+\frac{1}{9}\sin 2\theta+\frac{1}{27}\sin 3\theta+\cdots+\frac{1}{3^n}\sin n\theta$. Show that $C=\frac{3^{n+1}\cos\theta-3^n-3\cos((n+1)\theta)+\cos(n\theta)}{3^n(10-6\cos\theta)}$.

Question 19.

a) Show that $C = \frac{-\cos((2n+1)\theta)+\cos((2n-1)\theta)}{2-2\cos 2\theta}$.

b) Show that $S = \frac{2\sin\theta-\sin((2n+1)\theta)+\sin((2n-1)\theta)}{2-2\cos 2\theta}$.

Question 20. Use Binomial expansion to expand each of the following:

a) $(1+e^{i\theta})^4$

b) $(1+e^{i\theta})^5$

c) $(1+e^{i\theta})^n$

d) $(1+e^{2i\theta})^n$

Question 21.

a) Find $1 + \cos 2A$ in terms of $\cos A$.

b) State $\sin 2A$ in terms of $\sin A$ and $\cos A$.

c) Hence show that $1 + e^{i\theta} = 2e^{i\theta/2} \cos \frac{1}{2}\theta$.

Question 22. Finite series C and S are defined by $C = 1 + 4\cos\theta + 6\cos 2\theta + 4\cos 3\theta + \cos 4\theta$ and $S = 4\sin\theta + 6\sin 2\theta + 4\sin 3\theta + \sin 4\theta$.

a) Show that $1 + e^{i\theta} = 2e^{i\theta/2} \cos \frac{1}{2}\theta$.

b) Show that $C = 16\cos^4(\frac{1}{2}\theta)\cos(2\theta)$.

c) Hence, show that $S = 16\cos^4(\frac{1}{2}\theta)\sin(2\theta)$.

Question 23. Finite series C and S are defined by $C = 1 - 5\cos\theta + 10\cos 2\theta - \cdots - \cos 5\theta$ and $S = -5\sin\theta + 10\sin 2\theta - \cdots - \sin 5\theta$.

a) Show that $1 - e^{i\theta} = -2ie^{i\theta/2}\sin\frac{1}{2}\theta$.

b) Show that $C = 32\sin^5(\frac{1}{2}\theta)\sin(\frac{5}{2}\theta)$.

c) Hence, show that $S = -32\cos^5(\frac{1}{2}\theta)\cos(\frac{5}{2}\theta)$.

Question 24. Finite series C and S are defined by $C = 1 + 6\cos\theta + 15\cos 2\theta + \cdots + \cos 6\theta$ and $S = 6\sin\theta + 15\sin 2\theta + \cdots + \sin 6\theta$. It is known that $1 + e^{i\theta} = 2e^{i\theta/2}\cos\frac{1}{2}\theta$.

a) Show that $C = 64\cos^6(\frac{1}{2}\theta)\cos(3\theta)$.

b) Hence, show that $S = 64\cos^6(\frac{1}{2}\theta)\sin(3\theta)$.

Question 25. Finite series C and S are defined by $C = 1 + 6\cos 2\theta + 15\cos 4\theta + \cdots + \cos 12\theta$ and $S = 6\sin 2\theta + 15\sin 4\theta + \cdots + \sin 12\theta$. It is known that $1 + e^{i\theta} = 2e^{i\theta/2}\cos\frac{1}{2}\theta$.

a) Show that $C = 64\cos 6\theta\cos^6\theta$.

b) Hence, show that $S = 64\sin 6\theta\cos^6\theta$.

Question 26. Finite series C and S are defined by $C = 1 + \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + \cos n\theta$ and $S = \binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \cdots + \sin n\theta$. It is known that $1 + e^{i\theta} = 2e^{i\theta/2}\cos\frac{1}{2}\theta$.

a) Show that $C = 2^n\cos^n(\frac{1}{2}\theta)\cos(\frac{n}{2}\theta)$.

b) Hence, show that $S = 2^n\cos^n(\frac{1}{2}\theta)\sin(\frac{n}{2}\theta)$.

Question 27. Finite series C and S are defined by $C = 1+3\cos\theta+3\cos 2\theta+\cos 3\theta$ and $S = 3\sin\theta+3\sin 2\theta+\sin 3\theta$.

a) Show that $1 + e^{i\theta} = 2e^{i\theta/2}\cos(\frac{1}{2}\theta)$.

b) Show that $C = 8\cos^3(\frac{1}{2}\theta)\cos(\frac{3}{2}\theta)$.

c) Hence, show that $S = 8\cos^3(\frac{1}{2}\theta)\sin(\frac{3}{2}\theta)$.

Question 28.

a) Given that $|z| < 1$, write down the sum of the infinite series $1 + z + z^2 + z^3 + \cdots$.

b) Hence, given that $z = \frac{1}{3}(\cos\theta + i\sin\theta)$, show that $\frac{1}{3}\sin\theta + \frac{1}{9}\sin 2\theta + \frac{1}{27}\sin 3\theta + \cdots = \frac{3\sin\theta}{10-6\cos\theta}$.

c) Show that the sum cannot be purely imaginary, giving a reason for your answer.

Question 29.

a) Given that $|z| < 1$, write down the sum of the infinite series $z + z^2 + z^3 + \cdots$.

b) Hence, given that $z = \frac{1}{4}(\cos\theta + i\sin\theta)$, show that $\frac{1}{4}\cos\theta + \frac{1}{16}\cos 2\theta + \frac{1}{64}\cos 3\theta + \cdots = \frac{4\cos\theta - 1}{17 - 8\cos\theta}$.

c) Determine for which values of θ the sum is purely real. Give your answer in terms of π for any integer values.

Question 30. An Argand diagram shows points $z, z^2, z^3, \ldots$ where $z = \frac{1}{2}e^{i\theta}$.

a) What is the ratio of successive distances of the points from the origin?

b) What is the angle between successive position vectors on the Argand diagram?

c) Show that $\frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots = \frac{2\sin\theta}{5 - 4\cos\theta}$.

Question 31. Series C and S are defined by $C = \frac{1}{2}\cos\theta + \frac{1}{4}\cos 2\theta + \frac{1}{8}\cos 3\theta + \cdots + \frac{1}{2^n}\cos n\theta$ and $S = \frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots + \frac{1}{2^n}\sin n\theta$. Show that $S = \frac{2^{n+1}\sin\theta - 2\sin((n+1)\theta) + \sin(n\theta)}{2^n(5-4\cos\theta)}$.

Question 32. Given $S = \sum_{r=0}^{12} \binom{12}{r} \sin(r\theta)$, show that $S = 4096\sin(6\theta)\cos^{12}(\frac{1}{2}\theta)$.

Question 33.

$$C = 1 - \binom{16}{1}\cos\theta + \cdots + \cos 16\theta, \quad S = 1 - \binom{16}{1}\sin\theta + \cdots + \sin 16\theta.$$

Show that $C = 2^{16}\cos(8\theta)\sin^{16}(\frac{1}{2}\theta)$.

Question 34.

a) Show that $\frac{2}{1-e^{i2\theta}} = 1 + i\cot\theta$.

b) Finite series C and S are defined by $C = 2+2\cos\frac{\pi}{14}+2\cos\frac{2\pi}{14}+\cdots+2\cos\frac{6\pi}{14}$ and $S = 2(\sin\frac{\pi}{14}+\sin\frac{2\pi}{14}+\cdots+\sin\frac{6\pi}{14})$. Show that $C = 1 + \cot\frac{\pi}{28}$.

c) Hence show that $S = C - 2$.

Question 35. You are given that $C = \cos\theta + \cos(\theta + \frac{4\pi}{5}) + \cos(\theta + \frac{8\pi}{5}) + \cos(\theta + \frac{12\pi}{5}) + \cos(\theta + \frac{16\pi}{5})$ and $S = \sin\theta + \sin(\theta + \frac{4\pi}{5}) + \sin(\theta + \frac{8\pi}{5}) + \sin(\theta + \frac{12\pi}{5}) + \sin(\theta + \frac{16\pi}{5})$.

a) Show that $C + iS = e^{i\theta}\sum_{k=0}^{4}(e^{4\pi i/5})^k$.

b) Show that $C = 0$ and $S = 0$.

Question 36. Let $C = \cos 0 - \cos\frac{\pi}{2} + \cos\pi - \cos\frac{3\pi}{2}$ and $S = \sin 0 - \sin\frac{\pi}{2} + \sin\pi - \sin\frac{3\pi}{2}$.

a) Write $C + iS$ in summation notation.

b) Show that $C = 0$ and $S = 0$.

Question 37. The infinite series C and S are defined as $C = 1 + k\cos\theta + k^2\cos 2\theta + k^3\cos 3\theta + \cdots$ and $S = k\sin\theta + k^2\sin 2\theta + k^3\sin 3\theta + \cdots$, where k is a real number and $|k| < 1$.

a) Show that $C = \frac{1-k\cos\theta}{1-2k\cos\theta+k^2}$.

b) Find a corresponding expression for S.

Question 38. The infinite series C and S are defined as $C = k\cos\theta + k^2\cos 2\theta + k^3\cos 3\theta + \cdots$ and $S = k\sin\theta + k^2\sin 2\theta + k^3\sin 3\theta + \cdots$, where k is a real number and $|k| < 1$.

a) Show that $C = \frac{k\cos\theta-k^2}{1-2k\cos\theta+k^2}$.

b) Find a corresponding expression for S.

Question 39. Finite series C and S are defined by $C = 1 + \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + \binom{n}{n}\cos n\theta$ and $S = \binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \cdots + \binom{n}{n}\sin n\theta$.

a) Show that $C = 2^n \cos^n(\frac{\theta}{2})\cos(\frac{n\theta}{2})$.

b) Find a corresponding expression for S.

c) Hence, show that $\frac{C}{S} = \cot(\frac{n\theta}{2})$.

Question 40. Finite series C and S are defined by $C = 1 - \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta - \cdots + (-1)^n \cos n\theta$ and $S = -\binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta - \cdots + (-1)^n \sin n\theta$.

a) Show that $1 - e^{i\theta} = -2ie^{i\theta/2}\sin\frac{\theta}{2}$.

b) Show that $C + iS = \left[-2i\sin\frac{\theta}{2}(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})\right]^n$.

c) Hence, show that, for even values of n, $\frac{S}{C} = \tan(\frac{n\theta}{2})$.

Question 41. It is given that $\int e^{(a+bi)x}\,dx = \frac{1}{a+bi}e^{(a+bi)x} + c$ for real values a and b. The integrals C and S are defined by $C = \int_0^{\pi/3} e^{3x}\cos 5x\,dx$ and $S = \int_0^{\pi/3} e^{3x}\sin 5x\,dx$.

a) By considering $C + iS$, show that $C = \frac{e^{\pi}}{68}(3 - 5\sqrt{3}) - \frac{3}{34}$.

b) Obtain a similar expression for S.

Question 42. The finite series C and S are defined as $C = 1 + \cos\theta + \cos 2\theta + \cos 3\theta + \cos 4\theta$ and $S = \sin\theta + \sin 2\theta + \sin 3\theta + \sin 4\theta$.

a) Show that $C + iS = \frac{e^{5i\theta/2} - e^{-5i\theta/2}}{e^{i\theta/2} - e^{-i\theta/2}} \times e^{2i\theta}$.

b) Hence show that $C = \frac{1}{2}\sin(4\theta)\cot(\frac{1}{2}\theta) + \cos^2(2\theta)$.

c) Determine the values of θ in the range $0 < \theta < \pi$ for which $C = S$, giving your answer in exact form.

Question 43. It is given that $P = e^{i\theta} + e^{2i\theta} + e^{3i\theta} + e^{4i\theta}$.

a) State the value of P for $\theta = 2n\pi$, where n is an integer.

b) Show that for $\theta \neq 2n\pi$ for integer n, $P = \frac{e^{i\theta/2}(e^{4i\theta}-1)}{2i\sin(\frac{1}{2}\theta)}$.

c) Show that for $\theta \neq 2n\pi$ for integer n, $\cos\theta + \cos 2\theta + \cos 3\theta + \cos 4\theta = \frac{\sin(\frac{9}{2}\theta)}{2\sin(\frac{1}{2}\theta)} - \frac{1}{2}$.

d) Hence show that $\theta = \frac{\pi}{5}$ is a root of $\cos\theta + \cos 2\theta + \cos 3\theta + \cos 4\theta = 0$ and find another root in the interval $0 < \theta \leq \frac{\pi}{2}$.